

Agentic AI Gmail Auto-Responder

Autonomous Multi-Agent Email Triage & Safe Draft Generation

An AI-driven system that scans real Gmail unread messages, analyzes actionable requests, performs live web research, and automatically creates professional response drafts directly inside the user's Gmail inbox.

Team

- 1) SHRIRAM N - 231501154
- 2) SANJAY KISHORE S.A - 231501145
- 3) THILLAI NATHAN B - 231501173

Framework

CrewAI + Groq Cloud — 100% free-tier,
on-demand execution

Safety First

Human-in-the-loop: zero auto-send,
drafts-only architecture

Problem Statement & Motivation

The Core Problem

Student and professional inboxes are overwhelmed with high-signal emails — placement notifications, hackathon invites, faculty announcements — that get buried in clutter. Missing these means missed opportunities.

Existing AI email bots compound the risk: they auto-send unverified replies, creating serious academic and professional liability.

1 High Volume of Critical Emails

Placement notifications, hackathons, and faculty announcements get lost in clutter — manual triage is unsustainable at scale.

2 Risk of AI Hallucinations / Auto-Sending

Standard AI bots auto-send unverified replies, creating academic and professional risks that are unacceptable in high-stakes environments.

3 Cost & Latency Constraints

Commercial API setups like OpenAI GPT-4 are expensive for daily use — a free-tier, high-throughput alternative is essential.

📌 Build a safe, **100% free-tier**, on-demand AI system that assists email management without taking control away from the user.

Architecture & Workflow

Architecture of the multi-agent crew flow

A sequential pipeline from CLI trigger to final Gmail draft creation



1

Trigger

User initiates the process on-demand.



2

Authentication

Validates access via token.json.



3

Ingestion

The Gmail API queries unread messages using is:unread filters.



4

Execution

A sequential CrewAI crew analyzes, researches, and writes.



5

Output

The system pushes the final response directly to the Gmail Drafts folder.

Technology Stack

Production-Grade, 100% Free-Tier Engineering Stack

Multi-Agent Framework

CrewAI v1.15.17 with CrewAI Flow for state management and event listeners. Enables sequential agent orchestration with shared context across the pipeline.

LLM Engine

Groq Cloud Platform running open-source `openai/gpt-oss-20b`. Zero cost, ultra-high throughput, with native tool calling support — no OpenAI billing required.

Search & Research Tools

SerperDev API (Google SERP) and **Tavily Search API** for real-time web verification and fact-gathering during agent execution.

Email Integration

Official Google Gmail API with OAuth 2.0 via `google-auth-oauthlib` and `google-api-python-client`. Granular scopes, no plain-text password storage.

Language & Environment

Python 3.10 in a virtual environment (`.venv`). Pydantic for state typing, `dotenv` for secure secrets management, UTF-8 encoding resilience for Windows.

Multi-Agent Crew Architecture

3 Core Agents — Sequential Execution with Shared State



Agent 1: Email Filter Agent

Role: Ingestion & Categorization

Scans unread emails, parses sender, subject, and snippet metadata, and categorizes each message by urgency:

Action Required, FYI, or Spam.



Agent 2: Action Analyzer Agent

Role: Context & Fact Research

Identifies queries and deadlines embedded in emails, then uses live web search tools to gather background facts such as contest rules, registration deadlines, and event details.



Agent 3: Response Writer Agent

Role: Draft Synthesis & API Integration

Synthesizes professional, context-aware replies and invokes the custom `create_draft` tool to publish finalized drafts directly to Gmail Drafts.

Custom Tool Implementation

Custom Gmail Tool — `create_draft`

The core integration tool that bridges the AI agent layer with the Gmail API. Handles the full email construction and publishing pipeline:

- Uses `MIMEText` & `base64` encoding to build RFC 2822-compliant email structures
- Interacts directly with the `service.users().drafts().create()` endpoint
- Extracts clean recipient addresses from raw email headers
- Automatically appends "Re:" subject prefixes for proper threading

Payload-Optimized Web Search Tool — `web_search`

A purpose-built wrapper around the SerperDev REST API, engineered specifically for free-tier LLM constraints:

- Returns trimmed **250-character snippets** per search result
- Prevents token-per-minute (TPM) rate-limit violations on free LLM tiers
- Ensures agent research stays within Groq's throughput envelope without sacrificing factual accuracy

❏ Snippet truncation is a deliberate engineering decision — not a limitation — to maximize agent throughput on zero-cost infrastructure.

Key Safety & Security Features

Security-by-Design: Every Layer Engineered for Safety



Drafts-Only Architecture

The system **NEVER sends emails automatically**. It exclusively populates the Drafts tab — the user retains 100% send authority at all times.



On-Demand CLI Execution

Operates **only when manually triggered** via CLI. No background daemons, no scheduled tasks, no silent background consumption of credentials.



OAuth 2.0 Scopes Security

Uses granular Google scopes: `gmail.readonly`, `gmail.compose`, `gmail.modify`. No plain-text passwords stored anywhere in the system.



Rate-Limit & Encoding Resilience

Handles Windows UTF-8 encoding edge cases and implements **LLM retry logic** for seamless, error-free execution across environments.

Human-in-the-Loop Safety Mechanism

The Core Safety Differentiator — Control Stays With the Human, Always

Why It Matters

- **Zero Auto-Send Risk:** All generated drafts require explicit user review before any email leaves the inbox — no exceptions, no overrides.
- **Quality Control:** Users can edit, refine, or reject AI-generated responses before sending — the AI is an assistant, not a decision-maker.
- **Contextual Awareness:** Humans retain final decision-making authority over all outgoing communications, preserving professional and academic integrity.

Implementation Details

01

Draft Saved to Gmail

Agent publishes draft to Gmail Drafts folder via API — user must manually click "Send" in the Gmail UI.

02

Full Transparency

No background processes, no scheduled sends. Every action is visible, auditable, and reversible.

03

Future: Approval Dashboard

Interactive UI for batch review and one-click send — currently in roadmap as a future enhancement.

Future Enhancements

Roadmap: From MVP to Intelligent Personal Email OS

Smart Gmail Labeling & Auto-Tagging

Automatically apply custom Gmail labels — e.g., **AI-Actioned**, **Placement-Priority** — to processed inbox emails. Enables instant visual triage and searchable email history.

Interactive Web UI Approval Dashboard

Build a lightweight local **Streamlit** or **Next.js** UI to review, edit, or approve drafts with one click before pushing to Gmail. Replaces CLI with a visual control panel.

RAG & Personal Knowledge Memory

Integrate a local Vector DB (**ChromaDB**) containing student resume details, schedule, and FAQs. Enables fully personalized drafted responses without manual context injection.

Conclusion & Summary

Key Achievements

Built a **functional, agentic AI email assistant** integrated with the real Gmail API and free open-source LLMs — zero commercial API cost, production-grade reliability.

Impact

Saves significant time managing student and placement emails while guaranteeing **100% user control** and **zero risk** of accidental email sending through a drafts-only, human-in-the-loop architecture.

3 Agents

Filter → Analyze → Write

5-Stage Pipeline

Trigger → Auth → Ingest →
Execute → Draft

100% Free Tier

Groq + CrewAI + SerperDev

Zero Auto-Send

Human-in-the-loop always

Thank You! Questions & Discussion.